Domain-Routed Ensembles of Multi-Tier Supervised Detectors for Noise Event Detection in Crowd-Sourced Indic Speech

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***Abstract*—Crowd-sourced speech corpora recorded on mobile phones are a key resource for Indian-language speech technology, but they are full of non-speech noise events whose boundaries must be known to curate the data or to suppress the noise. We study noise event detection on the Vaani corpus in Track 1 of the IndoML 2026 Datathon, where an event counts as found only if both of its edges fall within 20% of its duration. An analysis of the data shows three obstacles: most training clips carry unverified or clip-level labels, events are short and closely spaced so that boundary errors rather than misses dominate, and the test set mixes natural recordings with synthesised ones whose statistics differ. We train 24 frame-level detectors on Whisper, WavLM and BEATs encoders with a multi-tier objective that uses every label tier together with explicit onset/offset heads, and route each clip to a domain-specific ensemble with its own decoding and boundary refinement. The system ranked 1st on the final leaderboard with Combined 1.6173 (Event F1 0.7895, Segment Dice 0.8278).**

***Index Terms*—sound event detection, weak supervision, boundary detection, domain routing, ensembles, Vaani**

I. Introduction

Speech technology for the many languages of India depends on large crowd-sourced corpora. Vaani [1] collects such speech on phones in more than 100 districts and therefore captures realistic acoustic conditions: traffic, animals, children, music and machines overlap or interrupt the speaker. These *noise events* differ from stationary background noise; they are short, varied and often louder than the speech. Knowing where they occur matters for filtering or relabelling training data, for analysing where recognisers fail, and for targeted noise removal, the subject of Track 2 of the same datathon [2].

Sound event detection (SED) [3] offers the natural formulation: predict the onset and offset of every event. Mature SED systems, however, are developed on curated benchmarks with a single label quality, moderate boundary tolerances and one recording domain. The Vaani noise event task departs from all three (Sec. II), which motivates our design.

Contributions. (1) An analysis of the task showing that boundary precision, not detection, limits performance, and that the test set contains two domains with different error profiles; (2) a multi-tier training objective that combines verified, unverified and clip-level labels with dedicated onset/offset heads, applied to 24 detectors with complementary front ends; (3) a learned domain router with branch-specific ensembles, decoding and boundary refinement; (4) a cross-fitted validation protocol for every tuning decision. *Final standing:* **1st place**, Combined **1.6173** (Event F1 0.7895, Segment Dice 0.8278; natural clips 0.5197/0.7256, synthetic clips 0.9981/0.9966), ahead of the second-ranked entry (1.6168).

II. Problem Analysis

Metric. Event F1 matches predictions one-to-one to reference events and accepts a match only if both edges lie within $\max(0.2\,d,\ 50\,\mathrm{ms})$ of the reference (d: its duration); Segment Dice measures overlap on a 10 ms grid per clip and is averaged over clips; the score is their sum. For a 0.3 s event the tolerance is 60 ms per edge, far tighter than the 200 ms collars common in SED evaluation [4]. Because Dice is averaged per clip, a clip without predictions scores 0, and deleting an overlapping prediction costs Dice even when it is an F1 false positive.

Labels. Of the 90,635 training clips only 12% carry verified timestamps; 68% have unverified timestamps and 20% clip-level labels only. The tiers carry different information and cannot simply be pooled.

Events. Events are short and dense: 44% of the reference events in natural validation clips last 0.25–0.5 s, human non-speech sounds (median 0.32 s) make up 43% of them, and 29% of the gaps between gold events are shorter than 0.3 s, so neighbouring events are easily merged.

Two domains. 38% of the test clips were synthesised from clean speech and pasted noise. Their sharp, exact edges make them easy once recognised (F1 0.99), whereas natural recordings remain hard (F1 0.5). Misrouting is costly in both directions (on validation, 0.06–0.09 Combined per misrouted clip), and an oracle router is worth +0.042, but a single noise-floor threshold identifies the domain for only 73.5% of validation clips and performs worse than no routing at all.

III. Related Work

Modern SED combines convolutional and recurrent layers [5] trained from strong and weak labels, with attention pooling turning frame scores into clip scores [6]; synthetic soundscapes with exact labels are a standard source of boundary supervision [7]. Large pretrained speech and audio encoders [8], [9], [10] with learned layer weighting [11] now provide the strongest front ends. These systems assume a single label quality and a single domain; we add tier-aware supervision and explicit domain routing.

IV. Training Data and Preprocessing

A. Vaani Noise Event Dataset

We train on the Vaani Noise Event Dataset [12] (ARTPARK-IISc/Vaani-Noise-Event-Dataset), derived from Vaani [1]: 90,635 clips (154.6 h) with 11,111 *gold* (verified timestamps), 61,640 *silver* (unverified timestamps) and 17,884 *bronze* clips (clip-level labels only); category strings are mapped to seven classes. Audio is decoded to 16 kHz mono, crops are at most 10 s and frame targets use a 20 ms grid (10 ms for two *hr* models). Two gold hold-out sets (1,111 and 3,999 clips) serve for model selection.

B. Official validation release

The 2,501 validation clips (1,806 natural, 695 synthetic with clean references) are the only additional data. Split into halves A/B (1,259/1,242 clips), they provide extra gold data for cross-fitted fold models (gold + one half, scored on the other) and three *full-gold* models (all 13,612 gold and validation clips), train the router and boundary regressor, and tune post-processing out-of-fold.

C. Boundary-exact augmentation

To obtain exact boundaries we synthesise 11,762 clips (21.4 h, 21,095 events) in the spirit of soundscape synthesis [7]: the 12,585 annotated event spans (4.0 h) and 5,884 event-free gaps (6.5 h) of 7,104 gold training clips form a noise and a clean bank; events are pasted at SNR $\sim \mathcal{U}(-22, -2)$ dB with 15 ms fades, and 35% of the inter-event gaps are drawn from 50–300 ms to mimic the dense events observed above.

V. Proposed Method

A. Frame-level detectors with multi-tier supervision

Each detector maps a waveform to 20 ms frame logits through five heads on a shared sequence representation: presence; onset and offset (Gaussian targets, $\sigma = 40$ ms, at gold boundaries); a 7-way category; and a clip-level weak head with linear-softmax attention pooling [6]. Front ends (Table I) are Whisper [8] encoders with positional embeddings sliced to the clip length and WavLM [9], each with learned layer weighting [11] and a 2-layer BiGRU; a hybrid that concatenates frozen BEATs [10] features (frequency-pooled, upsampled $\times 8$) with a Whisper-medium stream; and log-mel CRNNs [5]. The objective lets each tier contribute what it reliably provides:

$$\mathcal{L} = \mathcal{L}_{\text{gold}} + \lambda_s \mathcal{L}_{\text{silver}} + 0.3\,\mathcal{L}_{\text{cat}} + 0.5\,\mathcal{L}_{\text{weak}} + \mathcal{L}_{\text{on}} + \mathcal{L}_{\text{off}}, \quad (1)$$

where all terms are binary cross-entropies; presence terms use gold or silver timestamps ($\lambda_s = 0.25$, 0 for gold-only fine-tuning), boundary terms use gold clips only, and the weak term is the only signal from bronze clips. Base models are trained with AdamW [13] (head and encoder learning rates $5 \cdot 10^{-4}$–10^{-3} and $6 \cdot 10^{-6}$–$3 \cdot 10^{-5}$, encoder frozen for one epoch, bf16), then fine-tuned on gold data, on gold + one validation half, or on all gold + validation data.

TABLE I
The 24 detectors. Hugging Face revisions in parentheses. syn: all tiers + synthetic clips; split: all tiers minus a gold hold-out; FT: gold-only fine-tune; A/B: gold + one validation half; full: all gold + validation.

Encoder (revision)	Params	#	Training variants
Whisper large-v3 (06f233f)	640.6M	7	split, split+FT, syn, syn+FT, A, B, full
Whisper large-v2 (ae46427)	640.4M	2	split+FT, full
Whisper medium (abdf7c3)	310.4M	5	syn, A, B, syn and syn+FT at 10 ms
Whisper small (973afd2)	90.9M	3	syn, A, B
WavLM Base+ (4c66d48)	97.2M	1	split+FT
BEATs (172e407) + Whisper-m.	405.1M	2	split+FT, full
Log-mel CRNN	0.8/4.4M	4	split+FT, syn ($\times 2$), syn+FT

B. Domain routing and branch-specific decoding

Synthesised clips carry characteristic traces between events: a quiet, spectrally flat background, little high-band energy in quiet frames, a steady floor and energy jumps at splice points. A random forest [14] on 24 such statistics, trained on the `syntheticData` flags of the validation release, sends 2,080 of the 5,517 test clips to the synthetic branch. The *natural branch* averages 16 detectors with out-of-fold weights, blends in the full-gold models and two CRNNs, and decodes with hysteresis thresholds (0.57/0.48), event splitting at onset/offset peak pairs, duration-dependent edge snapping, a gradient-boosting [15] offset regressor, duplicate suppression and presence-based false-alarm rules. The *synthetic branch* averages 10 detectors with its own thresholds and snapping, as synthetic edges are sharp and its errors are mostly merged neighbours.

C. Refinement

For synthetic clips, our Track 2 enhancer [2] acts as a noise meter: the share of energy it removes over a predicted event has a median of -2 dB on true noise and -64 dB on noise-free speech, so 28 predictions with no other supporting evidence and a share below -12 dB are dropped. Because Segment Dice scores an empty clip as 0, each of the 31 natural clips without predictions receives one event over the ensemble's activity envelope (first to last frame with $p \geq 0.20$).

D. Use of validation-derived noise (disclosure)

The organisers' synthetic clips reuse noise recordings that also occur in the validation release, where subtracting the clean reference exposes them (1,181 distinct recordings). As approved in writing by the organisers, we locate these recordings in the test clips by FFT cross-correlation [16] and, where an energy test confirms the match, use the matched extent as a boundary check; on validation this extent equals the annotation for 1,925 of 1,926 copies with correlation ≥ 0.6. The check applies only to events whose noise recording also occurs in the validation release; all other events are determined by the learned detectors alone.

TABLE II
LEADERBOARD RESULTS ON THE 5,517 TEST CLIPS AS THE SYSTEM WAS BUILT UP.

System	Comb.	F1	Dice	$F1_{nat}$	$F1_{syn}$
Routed ensembles	1.5985	.7773	.8212	.5096	.9865
+ synthetic duration snapping	1.6001	.7788	.8213	.5096	.9892
+ post-processing, boundary check	1.6134	.7879	.8255	.5193	.9946
+ energy test, noise meter	1.6167	.7903	.8264	.5208	.9981
+ empty-clip handling (final, 1st)	**1.6173**	.7895	.8278	.5197	.9981

E. Implementation

PyTorch 2.11 [17], `transformers` [18] and scikit-learn [19] on NVIDIA H200 GPUs (141 GB; 71 GB MIG slices for fine-tuning); the 24 detectors process 10.9 h of audio in about 50 min. Code, configurations and all checkpoints: https://github.com/sajidjavid222/CodeAMU_Track1, https://huggingface.co/sajidjavid/CodeAMU-IndoML2026-Track1.

VI. RESULTS AND CONCLUSION

A. Protocol and main results

All decisions were taken on cross-fitted validation (fold models scored only on the unseen half; a change kept only if both halves improved), with the leaderboard as the final check. Domain routing is the decisive design choice: before it, our submissions scored 1.15–1.17 with synthetic-subset F1 below 0.38, while routed ensembles reach 1.60 (Table II). Boundary refinement then brings the synthetic subset close to its ceiling (F1 0.998, Dice 0.997). Individual detectors score 1.38–1.45 Combined on held-out validation, and the natural branch 1.173 out-of-fold on the 1,806 natural clips, so most of the advantage comes from the system built around the detectors.

B. Analysis

Natural clips remain the bottleneck (F1 0.52, against 0.998 on synthetic clips). Out of fold, 43% of reference events are matched, 46% are detected but miss the boundary tolerance (offsets more often than onsets) and only 10.5% are missed entirely. Edge errors of environmental sounds (vehicle, animal, baby) are scattered rather than biased (median 90–150 ms against 23–39 ms for human non-speech), which points to inherently ambiguous edges of gradual sounds rather than a correctable model bias: an oracle that fixed both edges of these near-misses would add +0.52 Combined, removing all pure false alarms only +0.02. The per-clip averaging of Dice explains two further observations: pruning overlapping predictions lowers the score, and giving every empty clip one event raises it.

C. Limitations and future work

The router learns the traces of one synthesis procedure: on differently generated clips (the Track 2 Phase 2 audio) the share of clips it could not assign confidently rose from 0.2% to 20.7%, and a second router built on detector outputs was needed. For natural recordings, boundary models that express uncertainty or learn annotator conventions are the most promising direction.

D. Conclusion

Tier-aware supervision, domain routing and branch-specific boundary refinement together gave the winning Track 1 system (1st place, Combined 1.6173). From cached detector outputs the released pipeline regenerates the final submission in 5,516 of 5,517 clips (one edge differs by 0.5 ms), and re-running four of the detectors reproduces their outputs bit for bit.

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